Chapter 4 · Semester 1

WORK, ENERGY AND POWER

Notes by Cikgu Suhaili

17 worked examples 2 simulations Exercises with answers

4.1 Work and energy

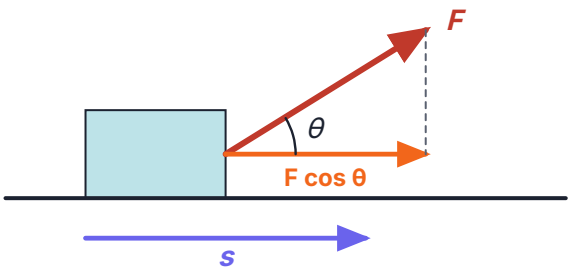
Work done by a constant force

Work done

is the **product of the component of the force parallel to the displacement and the magnitude of the displacement**.

$$W = \vec{F} \cdot \vec{s} = (F\cos\theta)s$$

F = magnitude of the constant force, s = magnitude of the displacement, θ = angle between **F** and **s**

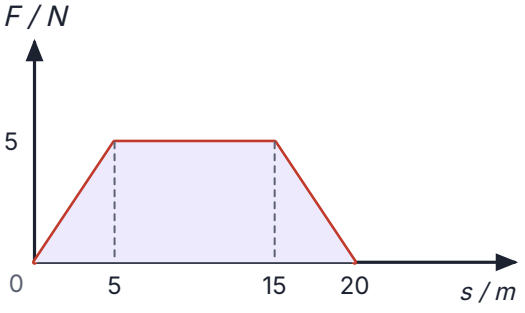


- Work is a **scalar**.
- Dimension: $[W] = [F][s] = \mathrm{M\,L^2\,T^{-2}}$.
- Unit: joule (J); 1 J = 1 N m = 1 kg m^2 s^{-2}.

Work done by a variable force

$$W = \int_{s_1}^{s_2} F\,\mathrm{d}s$$

Work done = area under the F–s graph.



Applications of W = F s cos θ for a constant force

Case	Angle	Work done
F in the same direction as **s**	θ = 0°	$W = Fs$
Two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ both along **s**	θ = 0°	$W = (F_1 + F_2)s = F_{\text{net}}s = ma\,s$
F perpendicular to **s**	θ = 90°	$W = 0$
F opposite to **s**	θ = 180°	$W = -Fs$

W = 0 when $F = 0$, or $s = 0$, or θ = 90°.

Example 4.1.1

A physics reference book is pushed 1.20 m along a horizontal table by a horizontal force of 3.00 N. The frictional force is 0.60 N. Calculate the work done by (a) the 3.00 N force, (b) the frictional force, (c) the normal force, and (d) the total work done on the book.

Solution

(a) $F = 3.00$ N, θ = 0°: $W = 3.00 \cos 0° \,(1.20)$ = **3.60 J**

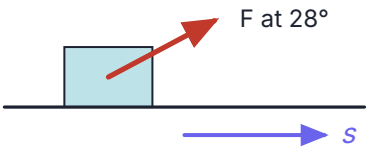
(b) $f = 0.60$ N, θ = 180°: $W = 0.60 \cos 180° \,(1.20)$ = **−0.72 J**

(c) R, θ = 90°: $W = R \cos 90° \,(1.20)$ = **0 J**

(d) $\Sigma W = 3.60 - 0.72 + 0$ = **2.88 J**, or $\Sigma W = (F - f) \cos 0° \; s = (3.00 - 0.60)(1.20) = 2.88$ J

Example 4.1.2

An object is pulled along the ground by a 75 N force directed 28° above the horizontal. How much work does the force do in pulling the object 8.0 m?

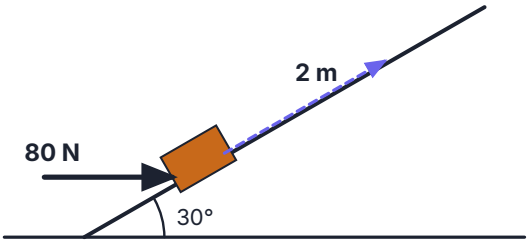


Solution

$$W = F \cos\theta \, s = 75 \cos 28° \, (8.0) = \mathbf{0.53\ kJ}$$

Example 4.1.3

A particle of mass 5 kg lies on a rough slope of angle 30°. A horizontal force of 80 N is applied to it. Find the work done by this force in pushing the particle 2 m up the slope.



Solution

The angle between the horizontal force and the displacement up the slope is 30°.

$$W_F = F \cos\theta \, s = 80 \cos 30° \, (2) = \mathbf{139\ J}$$

Exercises: work done by a constant force

1. A 300 g object slides 80 cm along a horizontal tabletop. How much work is done by friction if the coefficient of kinetic friction is 0.20?

Answer

−0.47 J

FULL SOLUTION

m = 0.300 kg, s = 0.80 m, μ_k = 0.20

f = $\mu_k$mg = 0.20(0.300)(9.81) = 0.589 N, opposite to the motion

W = fs cos 180° = −0.589 × 0.80

= **−0.47 J**

2. A force of (2**i** + 3**j**) N acts through a displacement of (4**i** + 5**j**) m. Find the work done.

Answer

23 J

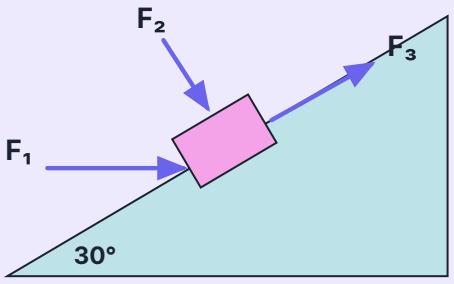
FULL SOLUTION

W = F · s = (2i + 3j) · (4i + 5j)

= 2(4) + 3(5) = 8 + 15

= **23 J**

3. A block moves 80 cm up a 30° incline under three forces: F_1 = 40 N (horizontal), F_2 = 20 N (perpendicular to the incline) and F_3 = 30 N (along the incline). Calculate the work done by each force.



Answer

W_1 = 28 J, W_2 = 0, W_3 = 24 J

FULL SOLUTION

s = 0.80 m along the incline.

F_1 is horizontal, so it makes 30° with the incline: $W_1 = F_1 s \cos 30° = 40(0.80)(0.866)$

W_1 = **28 J**

F_2 is perpendicular to the displacement: $W_2 = F_2 s \cos 90°$

W_2 = **0**

F_3 is along the incline: $W_3 = F_3 s = 30(0.80)$

W_3 = **24 J**

4. A 5.00 kg package slides 1.50 m down a ramp inclined at 12.0° below the horizontal; μ_k = 0.310. Calculate the work done on the package by (a) friction, (b) gravity, (c) the normal force, (d) the total work. (e) If its speed at the top is 2.20 m s^{-1}, what is its speed after sliding 1.50 m? (g = 9.81 m s^{-2})

Answer

(a) −22.3 J (b) 15.3 J (c) 0 (d) −7.0 J (e) 1.4 m s^{-1}

FULL SOLUTION

m = 5.00 kg, s = 1.50 m, θ = 12.0°, μ_k = 0.310

(a) N = mg cos 12.0° = 5.00(9.81)(0.978) = 47.98 N

f = μ_kN = 0.310 × 47.98 = 14.87 N

W_f = −fs = −14.87 × 1.50

= **−22.3 J**

(b) W_g = mg sin 12.0° × s = 5.00(9.81)(0.208)(1.50)

= **15.3 J**

(c) The normal force is perpendicular to the displacement: W_N = **0**

(d) W_{total} = −22.3 + 15.3 + 0

= **−7.0 J**

(e) Work–energy theorem: $W_{total} = ½mv^2 - ½mu^2$

$-7.01 = ½(5.00)v^2 - ½(5.00)(2.20)^2$

$v^2 = 4.84 - 2.80 = 2.04 \Rightarrow v =$ **1.4 m s^{-1}**

5. A 2.50 kg block is pushed 2.20 m along a frictionless horizontal table by a constant 16.0 N force directed 25.0° below the horizontal. Find the work done by (a) the applied force, (b) the normal force, (c) the gravitational force, (d) all the forces together.

Answer

(a) 31.9 J (b) 0 (c) 0 (both are perpendicular to the displacement) (d) 31.9 J

FULL SOLUTION

(a) W = Fs cos θ = 16.0(2.20) cos 25.0°

= **31.9 J**

(b) The normal force is perpendicular to the displacement: W = **0**

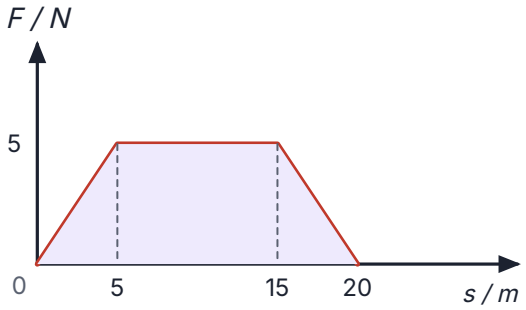
(c) Gravity is perpendicular to the displacement: W = **0**

(d) W_{total} = 31.9 + 0 + 0

= **31.9 J**

Example 4.1.4

Determine the work done on an object by the force F shown, if the object is displaced from s = 0 to s = 20 m.

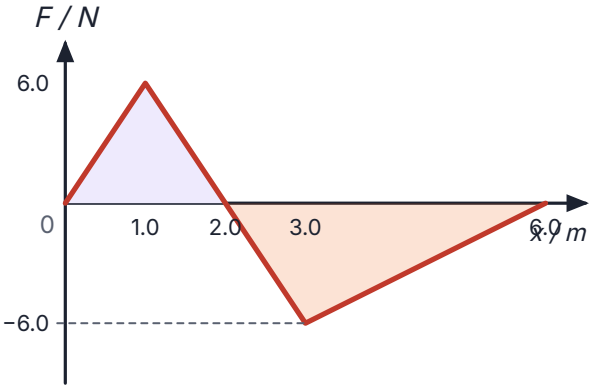


Solution

Work done = area under the graph = ½(5)(5) + (10)(5) + ½(5)(5) = **75 J**

Example 4.1.5

Calculate the work done by the variable force shown in the F–x graph.



Solution

Area above the axis (0 to 2.0 m) = ½(2.0)(6.0) = +6.0 J

Area below the axis (2.0 to 6.0 m) = ½(4.0)(−6.0) = −12 J

Total work = 6.0 − 12 = **−6 J**

4.2 Conservation of energy

Energy

is **the ability to do work**. Dimension $\mathrm{M\,L^2\,T^{-2}}$, unit joule (J), a scalar.

Two types of **mechanical energy**: potential energy, U, and kinetic energy, K. Two types of potential energy: gravitational, $U = mgh$, and elastic (spring), $U_s = \frac{1}{2}kx^2$.

Gravitational potential energy, U

Gravitational potential energy

is the **energy stored in a body or system because of its position**.

$$U = mgh$$

h = height of the body above the initial (reference) position

Elastic potential energy, U_s

Elastic potential energy

is the **energy stored in an elastic material as a result of stretching or compressing** it. Springs store elastic potential energy when compressed or extended.

Hooke's law

The restoring force, F_s, of a spring is **directly proportional to the extension or compression**, x, if the limit of proportionality is not exceeded.

$$F_s \propto -x \Rightarrow \boldsymbol{F_s = -kx}$$

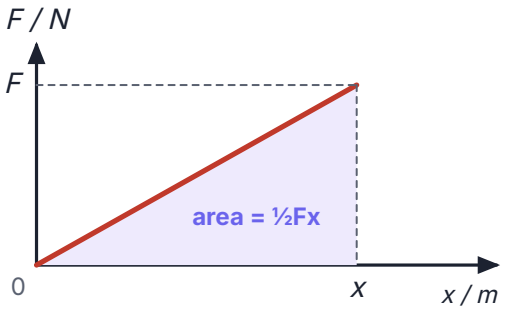
- k = spring constant (force constant); $x = x_f - x_i$.
- For calculation, if the object is stationary: $F_s = kx = F$ (the applied force).
- $[k] = \dfrac{[F_s]}{[x]} = \mathrm{M\,T^{-2}}$; unit kg $\mathrm{s^{-2}}$ or N $\mathrm{m^{-1}}$.

Work done by F to stretch the spring from its natural length by x: $W = \int_0^x F\,dx$, with $F = kx$

$W = \int_0^x kx\,dx = \frac{1}{2}kx^2$

$$U_s = \frac{1}{2}kx^2 = \frac{1}{2}F_s x$$

The work done equals the area under the straight-line F–x graph.



Example 4.2.1

A force of 600 N causes an extension of 10 cm in a spring. Find the elastic potential energy of the spring when (a) the extension is 12 cm, (b) a mass of 20 kg hangs vertically from it. (g = 9.81 m s^{-2})

Solution

$k = \frac{F}{x} = \frac{600}{0.1} = 6000$ N m^{-1}

(a) $U_s = \frac{1}{2}kx^2 = \frac{1}{2}(6000)(0.12)^2 =$ **43.2 J**

(b) $mg = kx \Rightarrow x = \frac{(20)(9.81)}{6000} = 3.27 \times 10^{-2}$ m; $U_s = \frac{1}{2}(6000)(3.27 \times 10^{-2})^2 =$ **3.21 J**

Exercises: springs

1. A 10 N force compresses a spring of spring constant 4.0 × 10^2 N m^{-1}. What is the compression?

Answer

2.5 cm

FULL SOLUTION

F = kx ⇒ x = F/k = 10 / (4.0 × 10^2)

x = 0.025 m = **2.5 cm**

2. It takes 400 J of work to stretch a spring by 8.00 cm. What is the spring constant?

Answer

1.25 × 10^5 N m^{-1}

FULL SOLUTION

$W = \frac{1}{2}kx^2$, $x = 8.00\text{ cm} = 0.0800\text{ m}$

$k = 2W/x^2 = 2(400) / (0.0800)^2$

$= \mathbf{1.25 \times 10^5\ N\ m^{-1}}$

Kinetic energy, K

Kinetic energy

is the energy of a body due to its motion.

$$K = \frac{1}{2}mv^2$$

Work–energy theorem

Work–energy theorem

The work done by the net force on an object (the total work done on it) equals the change in its kinetic energy.

$$W_{\text{total}} = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

A constant net force F acts on mass m through displacement s: $F = ma$

From $v^2 = u^2 + 2as$: $a = \dfrac{v^2 - u^2}{2s}$

$$F = m\frac{v^2 - u^2}{2s} \Rightarrow Fs = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \Rightarrow \mathbf{W = K_f - K_i = \Delta K}$$

Work done by a conservative force: W = −ΔU

The work done by a conservative force is equal to the negative of the change in potential energy.

- The work done by a **conservative** force (gravity, a spring) depends only on the initial and final positions; it is independent of the path.

- The work done against a **non-conservative** force (friction) depends on the path: the longer the path, the more work is done against friction.

When an object falls from h_1 to h_2, gravity does positive work and the potential energy decreases: $W_g = mg(h_1 - h_2) = U_i - U_f = -(U_f - U_i) = -\Delta U$

When an object is lifted, gravity does negative work and the potential energy increases: $W_g = -mg(h_2 - h_1) = -\Delta U$

For a spring: $W = -\Delta U_s = -(\tfrac{1}{2}kx_f^2 - \tfrac{1}{2}kx_i^2)$

Example 4.2.2

A 5.0 kg body is initially at rest on a horizontal frictionless surface. A 15 N force accelerates it to 12 m s^{-1}. Calculate (a) the distance travelled, (b) the work done by the force, (c) the final kinetic energy.

Solution

(a) $a = \dfrac{15}{5.0} = 3.0\ \text{m s}^{-2}$; $12^2 = 0 + 2(3.0)s \Rightarrow s =$ **24 m**

(b) $W = 15 \cos 0° \ (24) =$ **360 J**

(c) $K = \tfrac{1}{2}(5.0)(12)^2 =$ **360 J**, equal to the work done, as the work–energy theorem says

Example 4.2.3

How much work is needed for a 65 kg runner to accelerate from rest to 10.0 m s^{-1}?

Solution

$$W = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2 = \tfrac{1}{2}(65)(10.0)^2 - 0 = \textbf{3.25 kJ}$$

Example 4.2.4

A 4.1 kg box of books is lifted vertically from rest through 1.6 m by an upward force of 60.0 N. Find (a) the work done by the applied force, (b) the work done by gravity, (c) the final speed of the box.

Solution

(a) $W = 60.0 \cos 0° \, (1.6) =$ **96 J**

(b) $W = (4.1 \times 9.81) \cos 180° \, (1.6) =$ **−64 J**

(c) $W_{\text{total}} = 96 - 64 = 32 \text{ J} = \frac{1}{2}(4.1)v^2 - 0 \Rightarrow$ **$v = 3.9 \text{ m s}^{-1}$**

Exercises: work–energy theorem

1. Calculate the force required to stop a 1500 kg car travelling at 10 m s^{-1} within 500 m.

   Answer

   150 N

   FULL SOLUTION
   Work done by the stopping force = change in kinetic energy: $-Fs = 0 - \frac{1}{2}mu^2$
   $F = \frac{1}{2}mu^2/s = \frac{1}{2}(1500)(10)^2 / 500$
   = **150 N**

2. A 6.0 kg block initially at rest is pulled along a horizontal frictionless surface by a constant horizontal force of 12 N. Find its speed after it has moved 3.0 m.

   Answer

   3.5 m s^{-1}

   FULL SOLUTION
   $W = Fs = 12 \times 3.0 = 36$ J
   $W = \frac{1}{2}mv^2 - 0 \Rightarrow 36 = \frac{1}{2}(6.0)v^2$
   $v = \sqrt{12} =$ **3.5 m s^{-1}**

3. A boy exerts a force of 11.0 N at 29.0° above the horizontal on a 6.40 kg box. Find the work done by the boy and the final speed of the box after it moves 2.00 m, if it starts at 0.500 m s^{-1} and slides without friction.

   Answer

   19.2 J; 2.50 m s^{-1}

FULL SOLUTION

Work done by the boy: $W = Fs\cos\theta = 11.0(2.00)\cos 29.0°$

$= \mathbf{19.2\ J}$

No friction, so $W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$:

$19.24 = \frac{1}{2}(6.40)v^2 - \frac{1}{2}(6.40)(0.500)^2$

$v^2 = (19.24 + 0.80) / 3.20 = 6.26$

$v = \mathbf{2.50\ m\ s^{-1}}$

4. An object is released from rest at P on a smooth plane inclined at 30°. At Q its speed is 3.2 m s^{-1}. Calculate the length PQ. (g = 9.81 m s^{-2})

Answer

1.0 m

FULL SOLUTION

Smooth plane: the loss of potential energy equals the gain in kinetic energy.

$mg(PQ\sin 30°) = \frac{1}{2}mv^2$

$PQ = v^2 / (2g\sin 30°) = 3.2^2 / (2 \times 9.81 \times 0.5)$

$= \mathbf{1.0\ m}$

5. A 2.00 kg block of ice slides 0.750 m from rest down a frictionless plane inclined at 36.9° below the horizontal. What is its final speed?

Answer

2.97 m s^{-1}

FULL SOLUTION

Vertical drop: $h = 0.750\sin 36.9° = 0.450$ m

$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{(2gh)} = \sqrt{(2 \times 9.81 \times 0.450)}$

$= \mathbf{2.97\ m\ s^{-1}}$

Principle of conservation of energy

Principle of conservation of energy

In an **isolated (closed) system**, the **total energy of the system is constant**. Energy may change form or be transferred from one system to another, but the total remains the same.

$$\Sigma E_i = \Sigma E_f \quad \text{(no external force)}$$

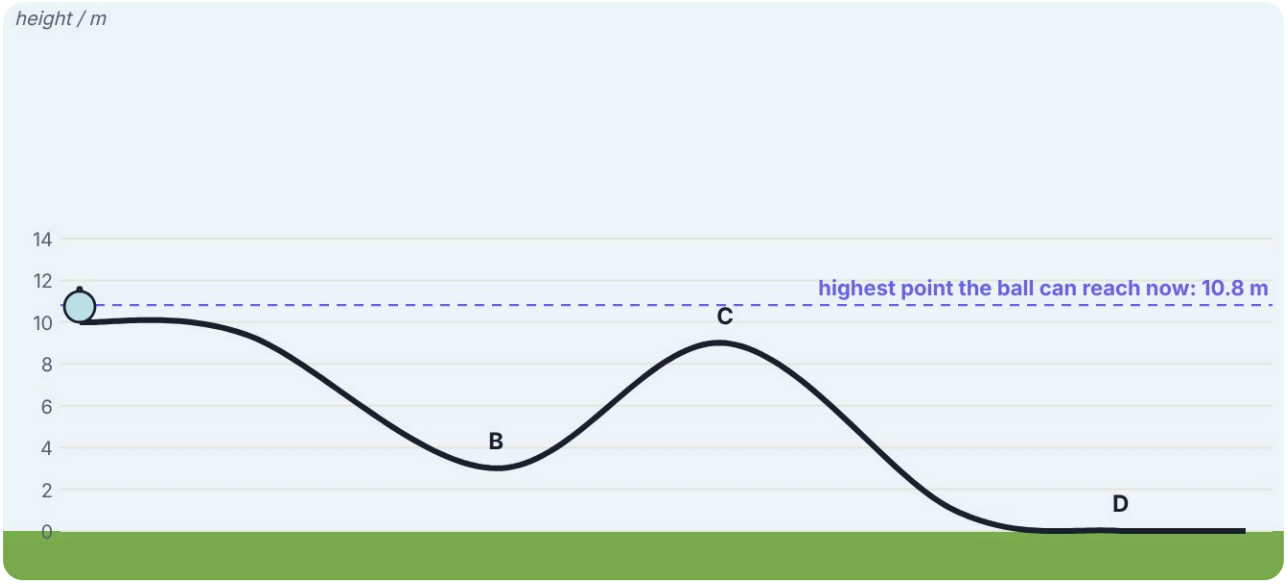
Without friction: $E = K + U$ = constant, so $K_i + U_i = K_f + U_f$

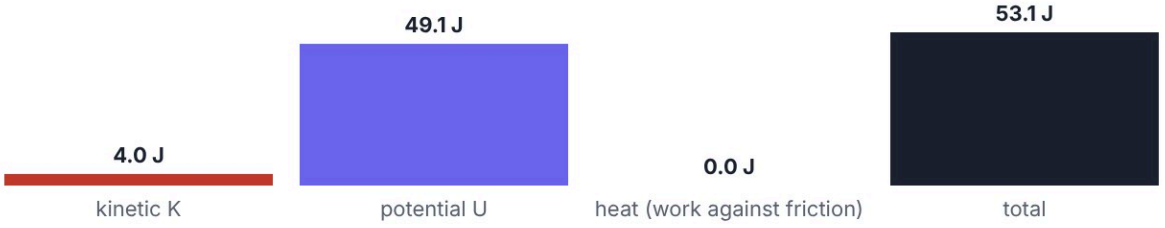
With friction, part of the mechanical energy becomes heat: $K_i + U_i = K_f + U_f +$ (work done against friction, fs).

Simulation

Energy on a track: kinetic, potential and heat

Release the ball from A and watch the energy bars. Without friction the total stays the same and the ball always reaches the same height. Turn on friction and the "heat" bar grows: that is the work done against friction.





speed	height	total energy at A	work done against friction
4.00 m/s	10.00 m	53.1 J	0.0 J

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 4 to use the simulation.

Example 4.2.5

A cricket ball of mass 0.156 kg is thrown upward with a kinetic energy of 45 J. What is the maximum height it can reach?

Solution

$\Sigma E_i = \Sigma E_f$: $K_i + U_i = K_f + U_f \Rightarrow 45 + 0 = 0 + mgh$

$$h = \frac{45}{(0.156)(9.81)} = \mathbf{29.4\ m}$$

Example 4.2.6

A 1.20 kg piece of cheese is placed on a vertical spring of negligible mass and force constant 1800 N m^{-1} that is compressed 15.0 cm. When the spring is released, how high does the cheese rise from its initial position? (The cheese and spring are not attached; g = 9.81 m s^{-2}.)

Solution

$K_i + U_i + U_{si} = K_f + U_f + U_{sf} \Rightarrow 0 + mgh_1 + \tfrac{1}{2}kx^2 = 0 + mgh_2 + 0$

$\tfrac{1}{2}kx^2 = mg(h_2 - h_1) = mg\Delta h$

$$\Delta h = \frac{\tfrac{1}{2}(1800)(0.15)^2}{(1.20)(9.81)} = \mathbf{1.72\ m}$$

A real example: the bungee jump

- From the platform to the point where the cord becomes tight, the jumper falls freely: gravitational potential energy → kinetic energy.
- After that, the cord stretches like a spring: kinetic and gravitational potential energy → **elastic potential energy** $\tfrac{1}{2}kx^2$.
- The jumper is fastest where the cord's pull equals the weight ($kx = mg$), not where the cord first tightens. At the lowest point the speed is zero and all the energy lost from gravitational PE is stored in the cord.

Simulation

Bungee jump: where does the energy go?

Change the jumper's mass and the stiffness of the cord, then jump. Watch the energy bars: the total stays the same (apart from a little lost to air resistance) while it moves between gravitational, kinetic and elastic potential energy.



fallen so far	speed	cord stretch x	lowest point: stretch from $mg(L+x) = \frac{1}{2}kx^2$
0.0 m	**0.0 m s^{-1}**	**0.0 m**	**14.2 m**

fastest where kx = mg, at stretch
2.94 m

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 4 to use the simulation.

Example 4.2.7

A 60 kg jumper uses a bungee cord of unstretched length 20 m and force constant 200 N m^{-1}. Ignoring air resistance, find (a) how far the cord stretches at the lowest point, (b) the jumper's greatest speed.

Solution

(a) Lowest point (speed zero): loss of GPE = elastic PE: $mg(20 + x) = \tfrac{1}{2}kx^2 \Rightarrow (60)(9.81)(20 + x) = 100x^2$

$100x^2 - 589x - 11\,772 = 0 \Rightarrow x =$ **14.2 m** (total fall 34.2 m)

(b) Greatest speed where $kx = mg$: $x = \dfrac{(60)(9.81)}{200} = 2.94$ m

$\tfrac{1}{2}mv^2 = mg(20 + 2.94) - \tfrac{1}{2}kx^2 = 13\,503 - 864 = 12\,639$ J $\Rightarrow v =$ **20.5 m s^{-1}**

Exercises: conservation of energy

1. A body of mass 5 kg slides from rest down a smooth slope. Find its speed after it has fallen a vertical distance of 10 m.

   Answer

   14 m s^{-1}

   FULL SOLUTION
   Smooth slope: mgh = ½mv^2 (the mass cancels)
   v = √(2gh) = √(2 × 9.81 × 10)
   = **14 m s^{-1}**

2. A smooth wire is bent into a circle of radius a held in a vertical plane. A bead threaded on the wire is released from rest at the highest point. Find its speed at the lowest point.

   Answer

   $\sqrt{4ag}$

   FULL SOLUTION
   The wire is smooth and the bead falls through the diameter, h = 2a.
   mg(2a) = ½mv^2
   v^2 = 4ag ⇒ v = **√(4ag)** (= 2√(ag))

3. A 1.70 kg block slides on a horizontal frictionless surface into a spring of force constant 955 N m^{-1}. The block stops after compressing the spring 4.60 cm. Find its initial speed.

Answer

1.09 m s^{-1}

FULL SOLUTION

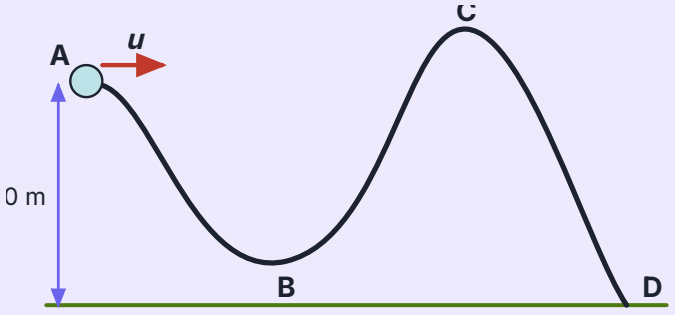
Kinetic energy of the block = elastic potential energy at maximum compression:

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow v = x\sqrt{(k/m)}$

$v = 0.0460 \times \sqrt{(955 / 1.70)}$

$= \mathbf{1.09\ m\ s^{-1}}$

4. A 0.50 kg ball is at A, 10 m high, with initial speed 4 m s^{-1} (no friction). Calculate (a) the total energy at A, (b) the speed at B, 3 m high, (c) the speed at D on the ground, (d) the maximum height of C so that the ball can pass over it.



Answer

(a) 53.1 J (b) 12.4 m s^{-1} (c) 14.6 m s^{-1} (d) 10.8 m

FULL SOLUTION

m = 0.50 kg, taking the ground as the zero of potential energy.

(a) $E_A = \frac{1}{2}mu^2 + mgh_A = \frac{1}{2}(0.50)(4)^2 + 0.50(9.81)(10)$

$E_A = 4.0 + 49.05 = \mathbf{53.1\ J}$

(b) At B (3 m): $\frac{1}{2}mv_B^2 = E_A - mgh_B = 53.05 - 0.50(9.81)(3) = 38.34$ J

$v_B = \sqrt{(2 \times 38.34 / 0.50)}$

$= \mathbf{12.4\ m\ s^{-1}}$

(c) At D (ground): $\frac{1}{2}mv_D^2 = 53.05$ J

$v_D = \sqrt{(2 \times 53.05 / 0.50)}$

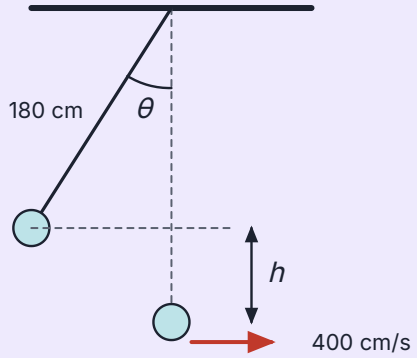
$= \mathbf{14.6\ m\ s^{-1}}$

(d) The ball just reaches the top of C when all its energy is potential: $mgh_C = 53.05$

$h_C = 53.05 / (0.50 \times 9.81)$

$= \mathbf{10.8\ m}$ (C must be lower than this)

5. A ball on the end of a 180 cm string swings as a pendulum. Its speed at the lowest position is 400 cm s^{-1}. (a) To what height h above this position does it rise before stopping? (b) What angle does the string then make with the vertical?



Answer

(a) 0.816 m (b) 56.9°

FULL SOLUTION

$u = 400\ \text{cm s}^{-1} = 4.00\ \text{m s}^{-1}$, $L = 180\ \text{cm} = 1.80\ \text{m}$

(a) $\tfrac{1}{2}mu^2 = mgh \Rightarrow h = u^2/(2g) = 4.00^2 / (2 \times 9.81)$

$= \mathbf{0.816\ m}$

(b) $\cos\theta = (L - h)/L = (1.80 - 0.816)/1.80 = 0.547$

$\theta = \mathbf{56.8°}$

Note: the notes give 56.9°, which comes from using $g = 9.8\ \text{m s}^{-2}$.

6. A 0.50 kg ball is thrown vertically upward with an initial kinetic energy of 80 J. Taking the launch point as the zero of potential energy, find (a) its kinetic and potential energies when it has travelled three-quarters of the way to its maximum height, (b) its speed there.

Answer

(a) $K = 20$ J, $U = 60$ J (b) 8.9 m s^{-1}

FULL SOLUTION

Total energy = 80 J (K = 80 J, U = 0 at the launch point). At maximum height $U_{max} = 80$ J.

(a) Three-quarters of the way up: $U = \tfrac{3}{4} \times 80$

$U = \mathbf{60\ J}$

$K = 80 - 60 = \mathbf{20\ J}$

(b) $\tfrac{1}{2}mv^2 = 20 \Rightarrow v = \sqrt{(2 \times 20 / 0.50)}$

$= \mathbf{8.9\ m\ s^{-1}}$

4.3 Power and mechanical efficiency

Power

is the **rate of doing work** or the **rate of energy transfer**.

$$P_{av} = \frac{\Delta W}{\Delta t} = \frac{\Delta E}{\Delta t}$$

Average power

$$P = \lim_{\Delta t \to 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

Instantaneous power

- Power is a **scalar**.
- $[P] = \frac{[\Delta W]}{[\Delta t]} = \frac{\mathrm{M\,L^2\,T^{-2}}}{\mathrm{T}} = \mathrm{M\,L^2\,T^{-3}}$; unit kg m^2 s^{-3} = J s^{-1} = watt (W).

$$P = \frac{dW}{dt} = \frac{(F\cos\theta)\,ds}{dt} \text{ and } \frac{ds}{dt} = v$$

$$P = Fv\cos\theta = \vec{F}\cdot\vec{v}$$

Example 4.3.1

A 1200 kg car can accelerate from rest to 25 m s^{-1} in 8.0 s. What average power must the motor produce? (Ignore friction.)

Solution

$$P_{av} = \frac{K_f - K_i}{\Delta t} = \frac{\tfrac{1}{2}(1200)(25)^2 - 0}{8.0} = \mathbf{47\ kW}$$

Example 4.3.2

An elevator of mass 1 Mg carries 10 passengers through a height of 15 m in 40 s. Calculate the average power required by the motor if no energy is lost. (g = 9.81 m s^{-2}; average mass per passenger 60 kg)

Solution

$M = 1000 + (60 \times 10) = 1600$ kg

Work done = change in potential energy = $mgh_f - mgh_i$, with $h_i = 0$

$$P = \frac{Mgh}{\Delta t} = \frac{(1600)(9.81)(15)}{40} = \mathbf{5886\ W}$$

Example 4.3.3

A 100 kg block is pulled at a constant 5.0 m s^{-1} across a horizontal floor by a 122 N force at 37° above the horizontal. At what rate does the force do work on the block?

Solution

$$P = Fv \cos\theta = (122)(5.0)\cos 37° = \mathbf{487\ W}$$

Exercises: power

1. A 50 kg person runs 200 m up a straight road inclined at 20° in 50 s. Neglecting friction and air resistance, calculate (a) the work done, (b) the average power. (g = 9.81 m s^{-2})

   Answer

   (a) 3.36 × 10^4 J (b) 672 W

   FULL SOLUTION

   Vertical height gained: h = 200 sin 20° = 68.4 m

   (a) W = mgh = 50(9.81)(68.4)

   = **3.36 × 10^4 J**

   (b) P = W/t = 3.355 × 10^4 / 50

   = **671 W**

   Note: the notes give 672 W, from dividing the rounded 3.36 × 10^4 J by 50 s.

2. You push a 75 kg box across a floor where μ_k = 0.65 with a horizontal force. (a) How much power is needed to push it at 0.50 m s^{-1}? (b) How much work do you do in 35.0 s?

   Answer

   (a) 240 W (b) 8.4 kJ

   FULL SOLUTION

   Constant speed ⇒ the push equals the friction: F = $\mu_k$mg = 0.65(75)(9.81) = 478 N

   (a) P = Fv = 478 × 0.50

   = **240 W**

   (b) W = Pt = 239 × 35.0

   = 8.4 × 10^3 J = **8.4 kJ**

Mechanical efficiency

Efficiency, η

is the **ratio of the useful work done (output) to the energy input**.

$$\eta = \frac{W_{out}}{E_{in}} \times 100\% = \frac{P_{out}}{P_{in}} \times 100\%$$

Example 4.3.4

An electric motor with a 2.0 horsepower input drives a machine with an efficiency of 45%. What is the energy output of the machine per second? (1 hp = 746 W)

Solution

$$0.45 = \frac{P_{out}}{2.0 \times 746} \Rightarrow P_{out} = 6.71 \times 10^2 \text{ W, so } \mathbf{6.71 \times 10^2\ J} \text{ per second}$$

Example 4.3.5

A machine lifts a 10 kg load through 2.5 m. The machine needs a force of 150 N to move 2 m. Determine its efficiency.

Solution

$W_{in} = F \cos\theta\ s = 150 \cos 0° (2) = 300$ J

$W_{out} = mgh = (10)(9.81)(2.5) = 245$ J

$\eta = \frac{245}{300} \times 100\% = \mathbf{81.7\%}$

Exercises: efficiency

1. The motor of an electric drill with an efficiency of 80% has a power input of 600 W. How much useful work is done by the drill in 30 s?

Answer

1.4×10^4 J

FULL SOLUTION

Efficiency = useful output / input ⇒ useful power = 0.80 × 600 = 480 W

W = Pt = 480 × 30

= **1.4×10^4 J**

2. A hoist lifts a 3000 kg load through 8.00 m in 20.0 s. The power supplied to the engine is 18.0 hp. Calculate (a) the work output, (b) the power output and power input, (c) the efficiency of the engine and hoist. (1 hp = 746 W)

Answer

(a) 235 kJ (b) 11.8 kW; 13.4 kW (c) 87.7%

FULL SOLUTION

(a) W_{out} = mgh = 3000(9.81)(8.00)

= 2.35×10^5 J = **235 kJ**

(b) $P_{out} = W_{out}/t = 2.354 \times 10^5 / 20.0$

= **11.8 kW**

P_{in} = 18.0 × 746

= 1.34×10^4 W = **13.4 kW**

(c) Efficiency = P_{out}/P_{in} × 100% = 11 772 / 13 428 × 100%

= **87.7%**

Real-World Physics

REAL-WORLD PHYSICS

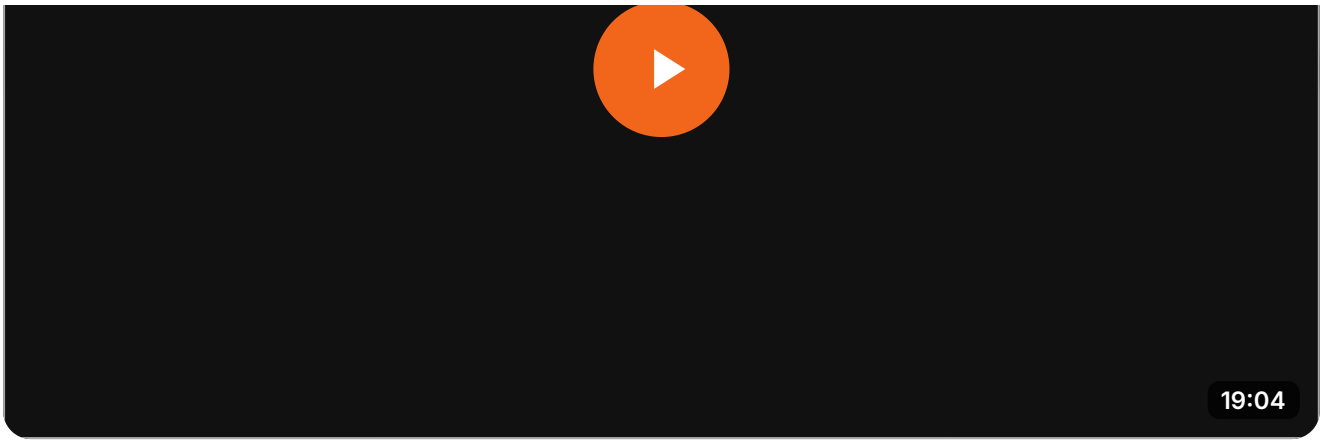
Langtang, Nepal · 26 August 2026

When a glacier falls 1200 m: energy on a mountain scale

On 26 August 2026 a large section of glacier ice and rock broke away from Langtang Lirung, north of Kathmandu, at about 5200 m and plunged about **1200 m** to the valley floor. It became a debris flow and flash flood that travelled almost 100 km down the Bhote Koshi–Trishuli valley, destroying roads, bridges and hydropower plants. More than 1400 people died.

The impact was so violent that seismometers recorded it like a magnitude 5.2 earthquake. Scientists estimate the falling mass at about **5×10^8 tonnes** of rock and ice.





Why Nepal's 2026 Flood Was Too Big for Any Single Defense · DT Terra

Open in YouTube ↗

While you watch: look for the moment the rock and ice accelerate down the slope, and notice how the debris carries enough energy and momentum to destroy bridges far downstream.

The physics

- Falling through height h, the mass loses gravitational potential energy mgh. The **work done by gravity** is converted mostly into kinetic energy.
- Work is also done **against friction** with the slope and the air, so part of the energy becomes internal energy (heat) and sound. This is why the real debris is slower than the "no friction" value.
- **Power** = rate of energy transfer. Releasing an enormous amount of energy in only minutes gives an enormous power.

Data used below: $m \approx 5.0 \times 10^{11}$ kg, $h \approx 1200$ m, $g = 9.81$ m s^{-2}. These are reported estimates, rounded for calculation.

Quick check 1

Calculate the loss of gravitational potential energy of the falling rock and ice.

A 5.9×10^{12} J

B 5.9×10^{14} J

C 5.9×10^{15} J

D 7.1×10^{18} J

Quick check 2

If all the potential energy lost became kinetic energy, what would be the speed of the debris at the valley floor?

A $49\ \text{m s}^{-1}$

B $110\ \text{m s}^{-1}$

C $153\ \text{m s}^{-1}$

D $2.35 \times 10^{4}\ \text{m s}^{-1}$

Quick check 3

The actual speed of the debris was much less than the value in Quick check 2. Which statement best explains this?

A Work is done against friction and air resistance, so part of the potential energy is converted into internal energy and sound.

B The total energy of the system decreases as the debris falls.

C Gravitational potential energy depends on the speed of the falling object.

D The kinetic energy of the debris is converted back into gravitational potential energy.

Quick check 4

Assume the energy in Quick check 1 is released over 2.0 minutes. What is the average power?

A $4.9 \times 10^{10}\ \text{W}$

B $4.9 \times 10^{13}\ \text{W}$

C $2.9 \times 10^{15}\ \text{W}$

D $7.1 \times 10^{17}\ \text{W}$

Sources: Reuters; BBC Verify; Al Jazeera; USGS (reports from August–October 2026). Video by an independent channel: preview before use in class. youtube.com/watch?v=Ss9fMd9C8os

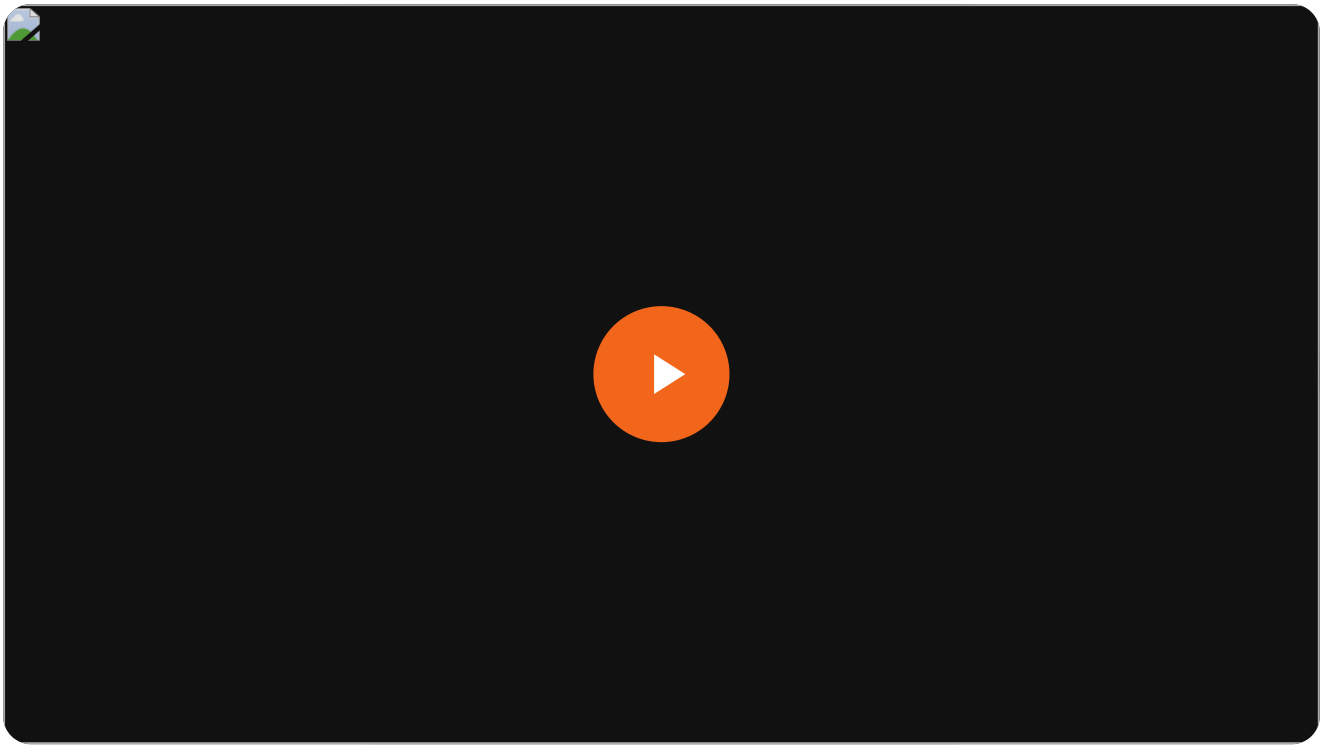
REAL-WORLD PHYSICS

Kölnbrein Dam, Austria · Richard Hammond's BIG

Inside a mega dam: how efficient is a hydroelectric turbine?

The Kölnbrein Dam in the Austrian Alps is a 200 m high concrete arch dam that holds back about 200 million tonnes of water. The water is piped down through the mountain, falling more than 1000 m to turbines in a power station far below.

In *Richard Hammond's BIG* (Discovery), Richard goes inside the dam and finds that its turbines convert about **92%** of the energy of the water into electrical energy.



How the Turbines in the Kölnbrein Dam are 92% Efficient · Richard Hammond's BIG

Open in YouTube ↗

While you watch: list every energy change from the water in the reservoir to the electricity leaving the power station.

The physics

- Water at height h has gravitational potential energy. As it falls through the pipe, this becomes kinetic energy.
- The fast water does **work** on the turbine blades and makes them rotate. The turbine turns a generator, which produces electrical energy by electromagnetic induction (Semester 2, Chapter 18).

- **Efficiency** $\eta = \frac{\text{useful output power}}{\text{input power}} \times 100\%$. The "missing" energy is not destroyed; it becomes internal energy (heat) and sound.

Quick check 1

Which order of energy changes is correct for a hydroelectric power station?

A Kinetic energy → gravitational potential energy → electrical energy

B Gravitational potential energy → kinetic energy of water → rotational kinetic energy of turbine → electrical energy

C Electrical energy → kinetic energy of water → gravitational potential energy

D Gravitational potential energy → electrical energy → kinetic energy of turbine

Quick check 2

Water flows at 20 $m^3\ s^{-1}$ and falls through a height of 1000 m to the turbines. The efficiency of the system is 92%. What is the electrical power output? (Density of water = 1000 $kg\ m^{-3}$)

A 1.80×10^5 W

B 1.96×10^8 W

C 1.80×10^8 W

D 2.13×10^8 W

Quick check 3

What happens to the remaining 8% of the input energy?

A It is destroyed inside the turbine.

B It is stored as gravitational potential energy in the dam wall.

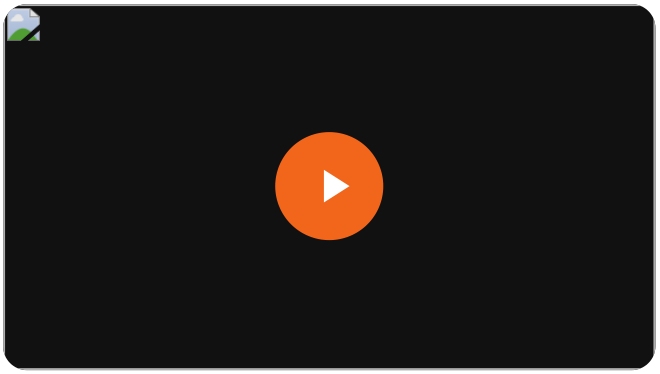
C It is converted into internal energy (heat) and sound by friction and turbulence.

D It is converted back into the mass of the water.

Source: *Richard Hammond's BIG*, Episode 3 "Austria's Mega Dam" (Discovery, 2020). Preview the clip before class. youtube.com/watch?v=DbvgK3PrSWw

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



REAL-WORLD APPLICATION

Calculating Mechanical Energy of a Roller Coaster

Open in YouTube ↗

Energy on a roller coaster

Gravitational potential energy changes into kinetic energy and back; the loss of mechanical energy equals the work done against friction and air resistance.